APEX RETAIL GROUP | PRACTICAL CLASS TASKS

DOCUMENT: CLASS TASKS (SESSION 6)

SCENARIO: PRODUCT CATEGORY PROFITABILITY & SEGMENTATION AUDIT

INSTRUCTIONS FOR LEARNER: Complete the following four tasks in your Jupyter Notebook during class. Follow the structured workflow: **Grouping Setup -> Multi-Metric Aggregation -> Sorting & Ranking -> Business Interpretation**. Do not peek at instructor solution keys until completed.

TASK 1: BASIC GROUPING & SUMMATION

Marcus Thorne (Product Category Manager) provides a clean transaction dataset:

```python
import pandas as pd

clean_transactions = {
    "transaction_id": ["TXN-701", "TXN-702", "TXN-703", "TXN-704", "TXN-705", "TXN-706", "TXN-707", "TXN-708", "TXN-709"],
    "category": ["Footwear", "Accessories", "Footwear", "Apparel", "Accessories", "Apparel", "Footwear", "Apparel", "Accessories"],
    "region": ["North", "South", "East", "North", "West", "South", "West", "East", "North"],
    "sales_amount": [450.00, 120.00, 890.00, 310.00, 95.00, 420.00, 620.00, 210.00, 150.00],
    "units": [3, 2, 6, 4, 1, 5, 5, 2, 2]
}
```

1. Import `pandas` as `pd` and load `clean_transactions` into DataFrame `df_txns`.
2. Use `.groupby()` grouped by `category` to calculate total `sales_amount` and total `units` sold per category.
3. Print the resulting grouped DataFrame.

TASK 2: MULTI-METRIC AGGREGATION & RENAME

1. Perform a multi-metric aggregation on `df_txns` grouped by `category`:
    - Calculate total sales (`sales_amount`: `sum`)
    - Calculate average sales per transaction (`sales_amount`: `mean`)
    - Calculate total units sold (`units`: `sum`)
    - Count total transactions (`transaction_id`: `count`)
2. Store the result in a DataFrame named `category_summary` and print it.

TASK 3: SORTING & FREQUENCY DISTRIBUTION

1. Sort `category_summary` in descending order based on total sales (`sales_amount`, `sum`). Print the sorted table.
2. Use `.value_counts()` on the `region` column to determine which region generates the highest transaction frequency.

TASK 4: EXECUTIVE SEGMENTATION MEMO

Review your sorted summary table and regional distribution counts:

1. Which product category generates the highest total sales volume?
2. Which region accounts for the highest transaction count?
3. Draft a 2-3 sentence strategic memo to Marcus Thorne summarizing category performance and recommending where to focus promotional inventory.